

Exact Posteriors for Values, Exact Costs for Choices: Two-Stage Learned Branching for Binary Linear Systems

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Abstract

A binary linear system (BLS) asks for $\mathbf{x} \in \{0, 1\}^n$ with $A\mathbf{x} = \mathbf{b}$, $A \in \{0, 1\}^{m \times n}$; on market-split-style instances the linear relaxation is uninformative and a complete search spends its effort on branching decisions. We present a branching heuristic learned in two stages from two exact signals, each answering a different question. Stage 1 answers *which value* a variable should take: on instances small enough to enumerate every solution, the exact conditional marginal $\Pr[x_j=1 \mid A, \mathbf{b}, \text{partial assignment}]$ is computable, and because conditioning a BLS on a partial assignment is identical to reducing it, one network trained on reduced instances predicts at every node of the tree. The marginals also partition a node’s variables into those on which a wrong branch empties the subtree and those on which it does not, and only the former can cost anything. Stage 2 answers *which variable* to branch on: the cost of a branching decision — the number of nodes its subtree expands — is produced exactly by the search itself, and an actor-critic fine-tuning on that cost, initialized at the Stage-1 rule with the marginal network frozen, changes only the branching order. On a 100-instance 21×60 set with matched wall-clock budgets and three seeds, Stage 1 is 2.0–2.3× faster than LP-guided branching with 5.5× fewer nodes; Stage 2 is a further 2.0–2.4× faster with 2.3–2.4× fewer nodes ($p < 0.001$), replicates across training seeds, and transfers from 21×60 to 18×50 . The fine-tuned policy branches on a different variable than the confidence rule at 95% of held-out states, typically one the network is *less* sure about; training Stage 1 on deployment-sized states instead, where its prediction margin lives, doubles the tree. The probability that a branch is right and the cost of it being wrong are different objectives, and the search minimizes the second. We report what the method does not do: CP-SAT remains 9–10× faster at 21×60 , and exact supervision is available only where enumeration completes.

Keywords: Binary linear systems, Learning to branch, Exact conditional marginals, Reinforcement learning, Tree search, Graph neural networks

1. Introduction

A binary linear system (BLS) is the feasibility problem

$$\text{find } \mathbf{x} \in \{0, 1\}^n \text{ such that } A\mathbf{x} = \mathbf{b}, \quad A \in \{0, 1\}^{m \times n}, \mathbf{b} \in \mathbb{Z}^m, \quad (1)$$

whose solution set we write $\mathcal{S}$. It arises in laser-based non-destructive testing, where a binary occupancy image is reconstructed from integer projections, and in its market-split form [5] it is a standard hard family: the relaxation polytope is large and its vertices are far from $\{0, 1\}^n$, so bounding prunes little and a complete search lives or dies by its branching decisions. Lattice reduction [1] solves many such instances outright and constraint solvers with clause learning solve the rest, but neither makes the branching decision itself less expensive to get right.

We ask what signal a learned branching heuristic for (1) should be trained on, and we find that the answer has two parts because a branching decision has two parts. *Which value* a chosen variable should try first is a question about where the solutions are, and it has an exact answer on instances small enough

to enumerate: the conditional marginal of x_j given the assignments made so far. *Which variable* to branch on is a question about the cost of the subtree that follows, and it has an exact answer too, produced by the search itself as it backtracks. The first signal is a probability; the second is a count. Our method trains on both, in that order, and the paper’s central empirical finding is that they are not interchangeable: improving the probability where it matters most does not shrink the tree, while optimizing the cost does, and the cost-optimal policy branches on variables the probability model is *less* sure about.

Contributions..

1. **An exact supervision signal for values, usable at every node.** Conditioning (1) on a partial assignment is identical to reducing it (Proposition 1), so one network trained on ordinary small instances predicts at the root and every interior node, and the exact conditional marginals both label it and partition each node’s variables into those that can cost a backtrack (FORCED) and those that cannot (OPEN). The FORCED share rises from 6% at the root to 88% at depth 8, which is why the network is trained in the tree.
2. **An exact cost signal for choices, read off the search.** An actor-critic fine-tuning of the branching order on per-decision subtree costs, initialized at the supervised rule

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with the marginal network frozen, expands 2.3–2.4× fewer nodes and is 2.0–2.4× faster than the supervised rule on 100 instances ($p < 0.001$, two training seeds), and transfers to a smaller size.

3. **Evidence that the two signals capture different things.** A residual measurement shows the confidence rule picks the cheapest of its five candidates only 22% of the time; training the marginal network on deployment-sized states raises its accuracy where its margin lives by one point and doubles the tree; the fine-tuned policy disagrees with the confidence rule at 95% of states.
4. **A controlled end-to-end evaluation,** with solution multiplicity matched across sizes, matched wall-clock budgets, warm-started relaxations on every arm, three seeds, a 100-instance test set, comparisons against CP-SAT and SCIP, and a deployed cascade with lattice reduction in front.

2. Related work

Complete solvers and lattice methods. Branch-and-bound over the relaxation gains little on market-split instances because the construction removes what bounding relies on [5]. Constraint solvers with conflict-driven clause learning [13] and lazy clause generation [14] manufacture information as they descend: a marginal uninformative at the root becomes 0 or 1 after a few commitments, which is why we study the interior of the tree. The Aardal–Hurkens–Lenstra embedding [1, 2] turns LLL/BKZ reduction [11, 16] into a certificate-producing first stage; we use it as such.

Learning to branch. Gasse et al. [7] imitate strong branching with a bipartite graph network, following Khalil et al. [10]; the gain is cheap access to an expert’s ranking, not new information, and our Stage 1 occupies the same structural position with an exact posterior in place of the expert. Imitation targets the expert’s ranking rather than the tree size the ranking is meant to reduce; Etheve et al. [6] and Scavuzzo et al. [15] optimize tree size directly, the latter formalizing branch-and-bound as a tree MDP in which a decision’s return is its subtree. Our Stage 2 uses that per-subtree credit with a plain policy gradient [19] and differs in its starting point and scope: it begins at a rule derived from exact marginals, freezes them, and fine-tunes the branching order only, in a feasibility search rather than an optimization. The classical statement of the objective is the fail-first principle [9]. Direct satisfiability prediction has a weaker record [17, 12], and message-passing networks provably cannot separate some feasible/infeasible pairs [4]; we never ask the network for a verdict.

Amortized inference. A learned map that replaces a per-instance computation by a single forward pass is amortized optimization [3, 8]. Both of our stages are of this kind: the marginal network amortizes LP-probing (one relaxation per variable) and the cost head amortizes the subtree costs that only a completed search reveals.

3. Method

3.1. Setting and search

Given (1), a depth-first search maintains a partial assignment $\mathbf{x}_F = \mathbf{v}$ on a set F of fixed variables and a set K of free ones. At each node it runs unit propagation on the residual system, prunes if a row becomes unsatisfiable, and otherwise branches: it selects $j \in K$ and a first value v , explores $x_j = v$ first and $x_j = 1 - v$ on backtracking. Variables are ever *fixed* only by sound deduction (propagation, and LP-probing where the opposite value makes the relaxation infeasible); the learned components decide the branching order and the first value, so a wrong prediction costs backtracking and nothing else. Figure 1 shows the deployed cascade: propagation, an LP-infeasibility rule, a kernel-pump heuristic and AHL/BKZ lattice reduction run first and close the easy part; the guided search sees the residual.

3.2. Conditioning is reduction; which decisions cost anything

Let $S(F, \mathbf{v}) = \{\mathbf{x} \in S : \mathbf{x}_F = \mathbf{v}\}$ be the solutions consistent with the node and $p_j = \Pr[x_j=1 \mid A, \mathbf{b}, \mathbf{x}_F=\mathbf{v}] = |\{\mathbf{x} \in S(F, \mathbf{v}) : x_j=1\}|/|S(F, \mathbf{v})|$ the conditional marginal, the posterior under the uniform prior on S that a planted-solution generator induces.

Proposition 1. *With A_F, A_K the column submatrices indexed by F, K , the map $\mathbf{x} \mapsto \mathbf{x}_K$ is a bijection from $S(F, \mathbf{v})$ onto $\{\mathbf{y} \in \{0, 1\}^{|K|} : A_K \mathbf{y} = \mathbf{b} - A_F \mathbf{v}\}$. Hence the conditional marginals of the original instance are the ordinary marginals of the reduced instance.*

Proof. $A\mathbf{x} = A_F \mathbf{x}_F + A_K \mathbf{x}_K$; with $\mathbf{x}_F = \mathbf{v}$ fixed, $A\mathbf{x} = \mathbf{b}$ iff $A_K \mathbf{x}_K = \mathbf{b} - A_F \mathbf{v}$, and $\mathbf{x}_F$ is determined, so the map is a bijection; the marginal statement follows by counting. $\square$

Two consequences (Figure 2). First, one architecture, one label generator and one training loop serve every node: a depth- d training state is stored as an instance on $n-d$ variables. Second, $p_j \in \{0, 1\}$ means every surviving solution agrees on x_j ; branching the other way empties the subtree. We call such variables **FORCED** and the rest **OPEN**: either branch of a **OPEN** variable still reaches a solution, so a wrong first value there costs nothing beyond exploring a subtree that contains one. Only **FORCED** errors cost a backtrack. Their share grows as the search descends and $|S(F, \mathbf{v})|$ shrinks — 6.3% at the root of a 10×25 instance, 87.9% at depth 8 (Section 4.2) — which is why prediction is trained and evaluated inside the tree.

3.3. Stage 1: exact conditional marginals as targets

On instances small enough to enumerate (10×25 here), S is obtained with the all-solutions mode of a constraint solver and p_j computed by counting. Only enumerations that run to exhaustion (**OPTIMAL** or **INFEASIBLE**) are accepted; a time-limited enumeration with solutions already collected would substitute a truncated S and bias every label. Training states are reachable in-tree states: a depth $d \in \{0, \dots, 12\}$ is drawn uniformly, a prefix of an LP-confidence ordering is fixed to the values of a randomly chosen actual solution, and the reduced

Deployed cascade (sound stages first; the learned components act only in the last stage)

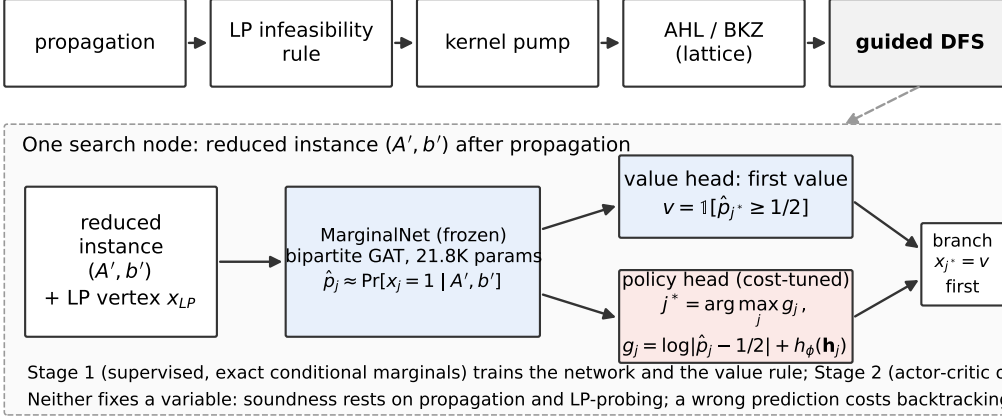
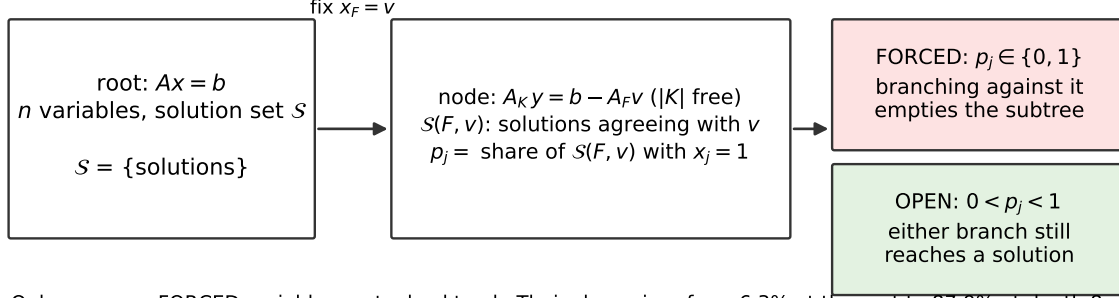


Figure 1: The deployed system and the two learned components. The cascade runs sound stages first; the learned components act only inside the final guided depth-first search. At a node, the frozen marginal network supplies $\hat{p}_j$; the value tried first comes from $\hat{p}$ (Stage 1), the variable branched on from the cost-tuned policy head (Stage 2). Neither fixes a variable.

Conditioning on a partial assignment is the same as reducing the instance (Proposition 1)



Only errors on FORCED variables cost a backtrack. Their share rises from 6.3% at the root to 87.9% at depth 8 of a 10×25 instance, so the same network is trained on in-tree states (reduced instances) rather than roots; the labels are exact because S is enumerated.

Figure 2: Proposition 1 and the partition it induces. A node with $\mathbf{x}_F = \mathbf{v}$ is the smaller BLS $A_K \mathbf{y} = \mathbf{b} - A_F \mathbf{v}$; its conditional marginals are the ordinary marginals of that instance. Variables on which every surviving solution agrees are FORCED; only branching against them costs a backtrack.

instance is enumerated in turn to give p_j on its free variables. The network f_θ (MarginalNet) is a bipartite variable–constraint graph network with graph attention [18]: three features per variable node (its LP value, column density, fractionality), three per constraint node (right-hand side, row density, LP residual, each normalized by row size), hidden width 64, four rounds of attention in each direction with residual connections and layer normalization, and one logit per variable; 21,761 parameters. It is trained by cross-entropy against the soft targets p_j ,

$$\mathcal{L}_1 = -\frac{1}{|K|} \sum_{j \in K} [p_j \log \hat{p}_j + (1 - p_j) \log(1 - \hat{p}_j)], \quad (2)$$

minimized exactly at $\hat{p} = p$. Replacing p_j by the planted solution, the usual choice, is a noisy surrogate whenever $|S| > 1$. At a node the network is evaluated once; the supervised branching rule is

$$j^* = \arg \max_{j \in K} |\hat{p}_j - \frac{1}{2}|, \quad v = \mathbb{I}[\hat{p}_{j^*} \geq \frac{1}{2}]. \quad (3)$$

3.4. Stage 2: exact subtree costs as the objective

Rule (3) maximizes the chance that the first child contains a solution. It says nothing about how much the search pays when it does not, nor how much smaller the children are; both are properties of the subtree, and a depth-first search produces them exactly. Let $c(u)$ be the number of nodes expanded below node u until a solution is found there or the subtree is exhausted; the search returns it when it backtracks out of u . Stage 2 fine-tunes the branching order on $-\log c(u)$.

Decision process. A state is the reduced instance after propagation; an action is a free variable j ; the first value stays $\mathbb{I}[\hat{p}_j \geq \frac{1}{2}]$ from the frozen Stage-1 network. Because each decision owns its subtree, credit is assigned per decision rather than per episode — the property that a sparse solved/unsolved reward lacks.

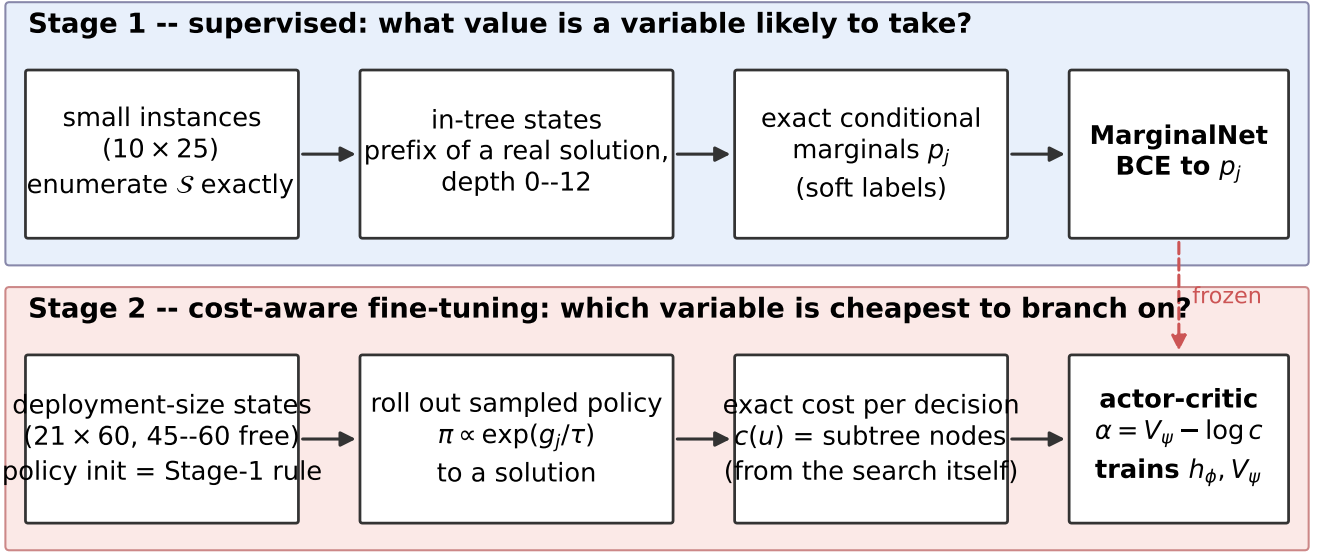


Figure 3: The two training stages. Stage 1 learns *which value* from exact conditional marginals of enumerable instances; Stage 2 learns *which variable* from the exact subtree cost of every decision, produced by the search, with the Stage-1 network frozen.

Policy and value.. On the frozen trunk of f_θ (variable embeddings $\mathbf{h}_j$) two heads are added. The policy logit is

$$g_j(s) = \log \left| \hat{p}_j - \frac{1}{2} \right| + h_\phi(\mathbf{h}_j), \quad (4)$$

with h_ϕ a two-layer network whose last layer is initialized to zero, so that $\pi(j | s) \propto \exp(g_j/\tau)$ starts as a softening of (3) and its argmax is exactly rule (3). A value head $V_\psi(s)$ on a pooled graph embedding, $\log |K|$ and $\log m$ predicts $\log c$.

Update.. Each epoch samples training states, rolls the sampled policy ($\tau = 1$) out to a solution, and records every decision with its cost. With the advantage $\alpha(u) = V_\psi(s_u) - \log c(u)$ standardized within the epoch,

$$\mathcal{L}_2 = \sum_u \left[-\alpha(u) \log \pi(j_u | s_u) - \lambda \mathcal{H}(\pi(\cdot | s_u)) + (V_\psi(s_u) - \log c(u))^2 \right], \quad \lambda = 0.01. \quad (5)$$

Only ϕ, ψ are trained; θ is fixed, so the marginal that chooses values and defines the FORCED/OPEN partition is unchanged, and any improvement is attributable to the branching order alone. The critic is pre-trained on decisions recorded under rule (3), and its held-out rank correlation with $\log c$ gates the actor update (a baseline that cannot rank costs makes the advantage noise). The epoch is selected on held-out states from the training pool, never on evaluation instances. At deployment the policy is greedy, $j^* = \arg \max_j g_j(s)$ (Figure 1).

3.5. Cost per node

At a node with n_c free variables the network costs one relaxation plus one forward pass; LP-probing, the sound procedure that detects the same FORCED variables, costs n_c relaxations.

With a warm-started relaxation on both sides the measured ratio grows from 1.7× at $n = 25$ to 7.2× at $n = 60$ and 17.8× at $n = 150$, but is only 1.3–1.7× at the depths where prediction matters on our instances, where probing additionally yields proofs. The Stage-2 head adds nothing measurable to the forward pass.

4. Experiments

4.1. Instances and protocol

Instances come from a market-split generator that plants a random $n/2$ -of- n solution and, among 20 candidate plantings per A , keeps the one with the largest relaxation-vertex spread; no uniqueness is imposed. Solution multiplicity changes the task — with $|S| = 1$ every variable is FORCED — so m is chosen per size to match it: 10×25 (median $|S| = 23$), 18×50 (19), 21×60 (10; 27/30 verified). At $n = 100$ a constraint solver finds no solution within 20 s, so search fails for every strategy and we stop at $n = 60$. Stage 1 trains on 10,000 in-tree states from 1,760 instances at 10×25 (Adam, 10^{-3} , 20 epochs, batch 1, gradient clipping 1.0; 20 minutes on one core); no hyperparameter was tuned and there is no validation split. Evaluation uses 30 instances per size, plus a 100-instance 21×60 set (generator seeds 700000–700099, containing the thirty) built after the 30-instance results were known. Stage 2 uses a separate 21×60 pool of 500 instances (seeds 800000+, 400/100 split), whose overlap with every evaluation set is zero by hash of $(A, \mathbf{b})$; its 100 held-out instances supply the states on which fine-tuning epochs are selected.

Every arm runs in the same loop: one warm-started relaxation per node (Gurobi dual simplex, bounds changed in place), identical propagation, identical value rule; arms differ only in

	root	depth 8	all depths
conditional Bayes ceiling	70.8%	94.4%	87.2%
MarginalNet (in-tree trained)	66.7%	84.3%	80.0%
MarginalNet (root trained, controlled)	—	77.4%	75.4%
LP relaxation rounding	60.9%	80.9%	76.5%
FORCED share of variables	6.3%	87.9%	—

Table 1: Per-variable accuracy on held-out in-tree states of the 10×25 pool against the exact posterior. The controlled row trains on the same pool, labels, architecture and state count, on root states only.

the branching choice. Budgets are per-instance wall-clock, enforced by process termination; concurrency is 6 on a 20-core machine; all timing-sensitive code is single-threaded. Per-instance speed is tested with a two-sided sign test; node counts are seed-invariant because the search is deterministic given the relaxation vertex. Every 21×60 figure is replicated over three seeds. Enumeration and CP-SAT use OR-Tools; lattice reduction uses fpylll.

4.2. Prediction against the exact posterior

Table 1 places the Stage-1 network against the ceiling that no predictor can exceed, $\text{ceil} = \frac{1}{|K|} \sum_j \max(p_j, 1 - p_j)$. It closes most of the gap between relaxation rounding and the ceiling at every depth, and its 3–6 point margin over rounding does not vanish deep in the tree. The composition, not the level, is what moves: the FORCED share rises from 6.3% to 87.9%, so almost every decision at depth 8 can cost a backtrack. Training on root states instead, with everything else controlled, gives a network that is *worse* than rounding inside the tree (75.4% against 76.4%): in-tree supervision is what makes learned guidance better than the classical rule. A graph-free MLP on the same features reaches 79.4% against the network’s 84.1% on a matched split, so the graph structure carries part of the margin. Where the margin lives at deployment size is shown in Figure 4c: inside the training range (13–25 free variables) network and rounding are indistinguishable (correlation 1.000), because at constraint density $m/|K| \geq 0.8$ the relaxation is already integral on every FORCED variable; the entire margin (+5 points) arises at 50–60 free variables, outside the training range, where the relaxation is weakest ($m/|K| \approx 0.35$ –0.42). 80% of the 21×60 search’s queries fall there.

4.3. Search performance

Table 2 is the main result. Three points. First, Stage 1 produces a smaller tree than the LP rule at every size (2.0×, 3.9×, 5.5–9.5× fewer nodes) and the ratio grows with size, as Figure 4c predicts; a smaller tree buys time only once the tree is large enough to pay for the forward pass, so at 10×25 Stage 1 is slower. Second, the 100-instance set replicates the 21×60 result with power — Stage 1 is faster on 70–71/100 instances — at smaller margins than the original thirty (2.0–2.3× against 2.6–3.0×), which are the first thirty seeds of the larger set and a favourable sample for Stage 1; we take the 100-instance figures as the estimate. Third, Stage 2 removes a further 2.3–2.4× of the nodes and 2.0–2.4× of the time on 100 instances, in both

training seeds and every evaluation seed, at unchanged per-node cost, for 4.8–5.9× over the LP rule end to end; the policy was fine-tuned at 21×60 only and transfers to 18×50 (1.7× fewer nodes, 1.8–2.2× faster). On the original thirty the two training seeds differ (one is near Stage 1, one 1.9× better), a spread the 100-instance set does not show. Random branching, for reference, solves 0/30 at 20×50 within 300 s and none of the 21×60 residual of Section 4.5.

4.4. Against complete solvers

The method does not compete with a complete solver (Table 3). The gap decomposes: at 21×60 CP-SAT’s own branch counter reports a median of 18,827 branches, more than our 4,040–8,873 nodes, while it processes 19,782 nodes/s against our ~ 700 in Python, where the network’s forward pass (1.9 ms at the root) is now over 90% of the per-node cost. The deficit is per-node throughput, not branching quality; whether it is recoverable with a compiled inference path and the clause learning and restarts our search lacks is not something we show.

4.5. Deployment in the full pipeline

Lattice coverage falls with size (30/30, 24–28/30, 10–16/30), which is what leaves room for branching to matter. At 21×60 the residual is genuinely hard — random branching closes none of the 50 residual instances pooled over seeds — and every sound strategy closes all of them; the difference is time. Stage 1 is 3.0× faster than the LP rule on the residual ($40/50$, $p = 2 \times 10^{-5}$) and Stage 2 a further 1.2–2.7× depending on the seed’s residual. End-to-end solve rate is 100% for every sound arm and 44% for random branching.

4.6. Size transfer of Stage 1

With $|S|$ matched, a Stage-1 network trained only at 10×25 is not worse at 21×60 than one trained at 18×50 : 8,657 against 14,742 median nodes, faster on 18–19/30 instances (n.s.). Root accuracy holds its margin over rounding at 10×25 , 20×50 and 40×100 (+11.0, +7.2, +4.7 points). This matters because exact labels exist only where enumeration completes; transfer lets a network trained there be deployed where they do not.

4.7. What the supervised target misses

Aligning the training distribution does not help. Figure 4c suggests an obvious fix: train Stage 1 where its margin lives. We built 2,691 in-tree states with 30–60 free variables from the 21×60 training pool (prefix of the LP ordering fixed to the planted solution; exact marginals; all states with ≤ 55 free variables enumerate within 90 s, 291/400 roots do) and trained the same architecture on them alone (M1) and on a half-and-half mix (M2). Accuracy on FORCED variables at 50–60 free variables rises by about one point (+6.1/ +10.7 against +5.4/ +9.7 for Stage 1). The tree doubles: M1 expands 21,337 median nodes on the 100-instance set against 13,386 (fewer on 38/100, $p = 0.021$), M2 18,558, and a switch that routes nodes with ≥ 50 free variables to M1 recovers nothing (17,252). The reason is where the nodes are: two thirds of all decisions occur at 25–34 free variables, and on such small dense states M1’s

size (budget)	strategy	solve rate	median nodes	median time	nodes/s	vs. previous row (per instance)
10 × 25 (60 s)	LP	100%	46	0.03 s	1,767	
	STAGE 1	100%	24	0.07 s	376	slower: forward pass exceeds a 24-node tree
18 × 50 (300 s)	LP	100%	4,224	2.43 ± 0.33 s	2,066	
	STAGE 1	100%	1,085	1.82 ± 0.37 s	731	1.3–1.4×, 18/30, n.s.
	STAGE 1+2	100%	424	0.52 ± 0.01 s	820	1.8–2.2×, 22–24/30, $p \leq 0.016$
21 × 60, 30 inst. (600 s)	LP	100%	82,366	47.9 ± 0.8 s	1,627	
	STAGE 1	100%	8,657	14.5 ± 0.1 s	619	2.6–3.0×, 23/30, $p = 0.005$
	STAGE 1+2 (seed 0 / 1)	100%	8,873 / 4,040	12.5 / 5.9 s	710 / 690	1.2–1.5× n.s. / 2.3×, 20–21/30
21 × 60, 100 inst. (600 s)	LP	100%	79,609	41.2 ± 1.9 s	1,966	
	STAGE 1	100%	13,386	16.5 ± 0.6 s	796	2.0–2.3×, 70–71/100, $p < 0.001$
	STAGE 1+2 (seed 0 / 1)	100%	5,495 / 4,940	7.6 / 6.8 s	738 / 730	2.0–2.2× / 1.9–2.4×, 64–68/100, $p \leq 0.007$

Table 2: Branching strategies in the same warm-started loop, lattice stage off, three seeds (mean ± s.d. of per-seed medians; node counts are seed-invariant). LP branches on the most integral relaxation value; STAGE 1 is rule (3); STAGE 1+2 the cost-fine-tuned policy, trained at 21 × 60 only and reported for two training seeds. The last column compares each row with the row above it.

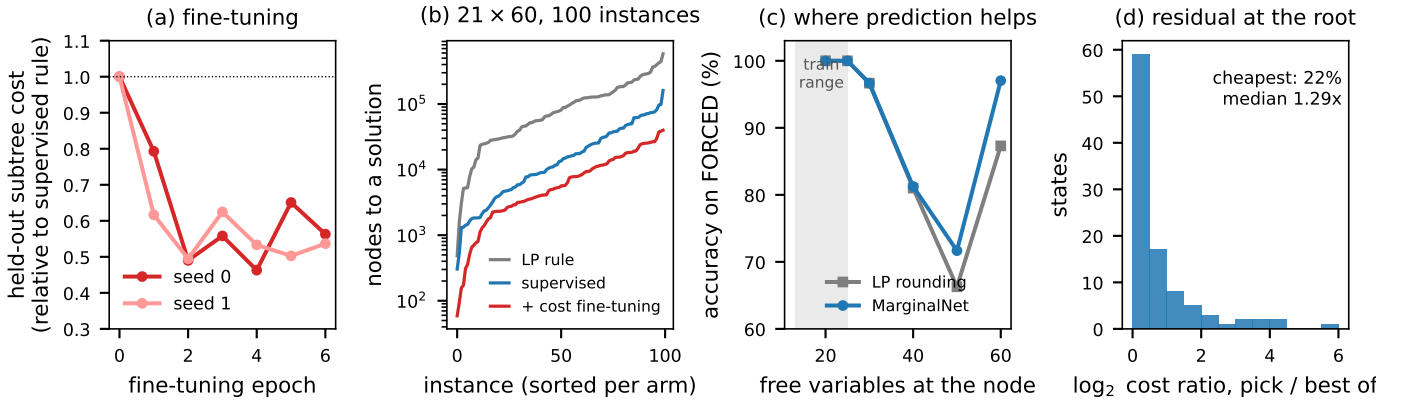


Figure 4: (a) Stage-2 fine-tuning: total subtree cost on 150 held-out states relative to the Stage-1 rule, two training seeds. (b) Nodes to a solution on the 100-instance 21 × 60 set, each arm sorted. (c) Accuracy on FORCED variables of 21 × 60 in-tree states by number of free variables: the Stage-1 margin over relaxation rounding lies outside the training range. (d) Residual at the root: cost of the confidence rule’s choice relative to the cheapest of its five candidates (100 held-out states, 45–50 free variables).

size	CP-SAT	SCIP	stage 1	stage 1+2
10 × 25	0.002 s	0.009 s	0.07 s	—
18 × 50	0.046 s	0.385 s	1.82 s	0.52 s
21 × 60 (30)	1.16 s	3.69 s	14.5 s	5.9–12.5 s
21 × 60 (100)	0.79 s	2.59 s	16.5 s	6.8–7.6 s

Table 3: Median solve time, single-threaded, same instances and budgets; every method solves every instance. CP-SAT is 9–11× faster than the full method at 21 × 60 and 11× at 18 × 50.

most confident variable is FORCED-and-correct 77.6% of the time against 94.4% for Stage 1 (rank correlation with the true margin 0.12 against 0.36). A network slightly better at the top of the tree and much worse inside it makes the search slower. Within 10 × 25, training on depths 0–6, 6–12 or 0–12 makes no difference (79.3/79.9/79.9%). The training-state distribution is not the lever.

The confident choice is rarely the cheapest.. On 100 held-out states with 45–50 free variables we forced the root to each of the five most confident variables and ran rule (3) below. The

size	strategy	lattice	residual	time on residual
10 × 25	any	30/30	—	—
18 × 50	LP / STAGE 1 / 1+2	24–28/30	11/11	0.5–5.0 / 0.6–2.5 / 0.8–0.8 s
21 × 60	random	10–16/30	0/50	> 600 s
21 × 60	LP	10–16/30	50/50	52.9–65.1 s
21 × 60	STAGE 1	10–16/30	50/50	14.2–20.2 s
21 × 60	STAGE 1+2	10–16/30	50/50	7.3–11.7 s

Table 4: Cascade with lattice reduction first, three seeds; lattice coverage varies with the seed’s column permutations, and every arm within a seed receives the identical residual. Median time on the residual instances, range over seeds.

rule’s own choice is the cheapest of the five on 22% of states; the median cost ratio to the best candidate is 1.29, the mean 2.97, a candidate at least twice as cheap exists on 24% of states, and the best root choice alone would remove 55% of the nodes summed over these states (Figure 4d). The residual is about *which* variable, and it is large.

4.8. Cost-aware fine-tuning

Critic gate.. Recording every decision of rule (3) on the training pool gives 37,275 decisions (9,375 held-out) with exact sub-

tree costs. The value head on the frozen trunk reaches a held-out Spearman correlation of 0.80 with $\log c$; a baseline that knows only the state’s size reaches 0.70, and within size buckets the critic’s advantage is larger (0.68 against 0.38 at 35–44 free variables). Unfreezing the trunk adds 0.02; the frozen critic is used.

Fine-tuning. Six epochs of 400 rollouts from the 45–60 free-variable training states ($\tau = 1$, about 21,000 decisions per epoch). Figure 4a shows the held-out monitor: total subtree cost falls to 0.46 (seed 0, epoch 4 selected) and 0.50 (seed 1, epoch 2) of the Stage-1 rule’s, and policy entropy from 1.7 to below 0.1. The evaluation is in Table 2 and Figure 4b.

What changed. On 300 held-out states the fine-tuned policy branches on a different root variable than rule (3) at 95% of them; its pick is typically the 15th most confident ($|\hat{p} - \frac{1}{2}|$ of 0.26 against 0.43) and is FORCED with a correct first value less often (57.5% against 82.5%). The policy moved away from the variables the network is surest about, and the search got cheaper. The gain is not a property of the root decision alone: forcing only the root to the fine-tuned pick with rule (3) below gives a median cost ratio of 0.93 (cheaper on 22/46), so it is the sequence of decisions that improves. Two training seeds reach similar costs while choosing different variables at many nodes, as expected if the objective, not a particular solution, was learned. We can state what the fine-tuning optimizes but not yet name the feature it exploits; a cheap proxy, the propagation the first child triggers, does not differ between the two choices.

5. Limitations

Exact supervision bounds the method. Marginals require enumerating $\mathcal{S}$, which does not complete beyond $n \approx 60$; Stage 1 transfers to larger sizes but cannot be trained there, and Stage 2’s cost signal, while available at any size the search runs at, was used at 21×60 only.

The useful size window is bounded on both sides. Below it lattice reduction closes everything; above it ($n = 100$) search fails for every strategy. Our demonstration lives at $n = 50$ –60.

The system is not competitive with CP-SAT. (9–11× slower), the deficit being per-node throughput in a Python loop dominated by the forward pass. Whether a compiled path plus clause learning closes it is not shown.

Statistical power and sampling. The original thirty 21×60 instances proved a favourable sample for Stage 1 and an unstable one for Stage 2; the 100-instance set is the estimate we stand behind, and Stage 2’s cross-size transfer rests on thirty instances at 18×50 .

Stage 2 rests on one training pool and two seeds. Its entropy collapses within four epochs, the epoch is selected by a monitor rather than a stopping rule, and the feature it exploits is unnamed. No hyperparameter of either stage was tuned; the depth range of Stage 1 was subsequently shown not to matter within a size.

Single instance family. Whether either signal transfers to other constraint families is untested.

6. Conclusion

A branching decision asks two questions, and we trained on an exact answer to each. The exact conditional posterior — computable where solutions can be enumerated, and transported to every node by the reduction identity — tells the search which value to try and identifies which predictions can cost anything; the exact subtree cost — produced by the search as it backtracks — tells it which variable to branch on. The first signal beats LP-guided branching by 2.0–2.3× on 100 hard 21×60 instances; the second, applied as a fine-tuning that leaves the first untouched, adds 2.0–2.4×, replicates across seeds and transfers down in size. That the fine-tuned policy prefers variables the posterior is less sure about, and that sharpening the posterior where its margin lives makes the tree larger, are the same fact seen twice: the probability of being right and the cost of being wrong are different objectives, and only the second is what a search minimizes.

Reproducibility

Code is organized under `proposed_src/` with a `ROLES.txt` per directory. Stage 1: `v15_bayes/v15_gen_by_m.py`, `v22_transfer/v22_split.py`, `v17_conditional/v17_make_conditional_data.py`, `v17_train_conditional.py`. Search and evaluation: `v23_ablation/run_revision2.sh`. Stage 2 and the residual experiments: `v24_deploy_range/run_v24.sh`, `run_v24_r1.sh`, `run_v24_r2.sh`, `run_v24_r2b.sh`; figures: `make_260917_figures.py`. All generators are seeded; instance data and weights are regenerated rather than distributed.

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